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Engineering
DIGITAL TWINS

IRT JULES VERNE
LE FUTUR DE VOS USINES

Digital twins of industrial robots manipulating soft physical objects.

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Physical twin description

One or several robot arms automating an industrial process requiring manipulating physical soft objects e.g., composite tissues or cables.

Digital twin description

At project start, the DT consists of an existing physical robotics platform, and preliminary models of the digital twin (incl. urdf and sysmlV2 models).

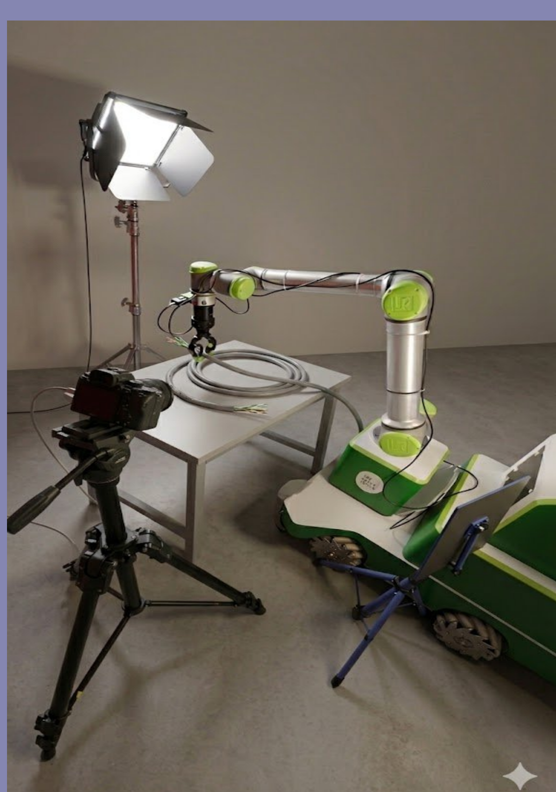
Digital twin usages

- *At design-time:* (i) Mastering complex robot behaviours, e.g., reachability, singular position and collision ; (ii) Foster collective work required for performing safety analysis of robotics systems (incl. SoS) for CE certification.
- *At run-time:* (i) Support for the cognitive behaviour of Robots to work in complex open environments, possibly including humans ; (ii) Enable robots to analyse and select the best set of actions to perform, manipulating physical soft objects, e.g., composite tissues or cables.

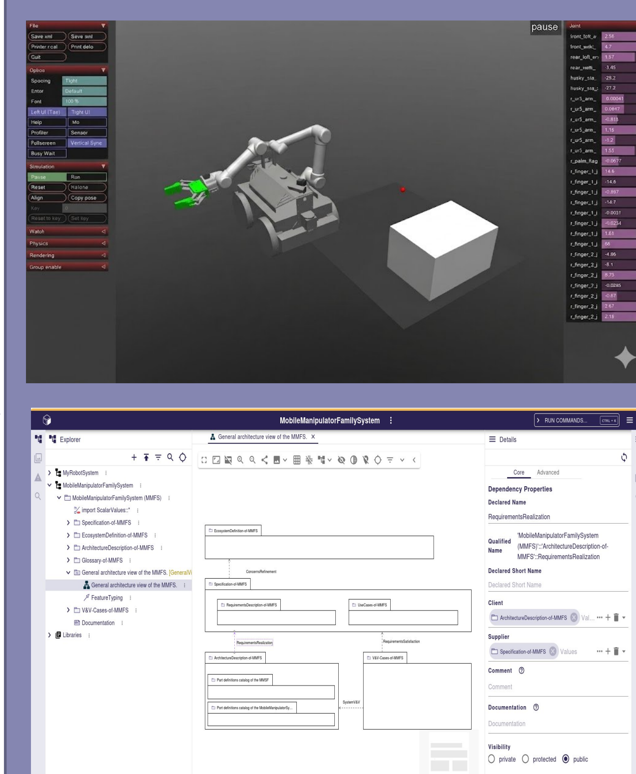
Engineering digital twin challenges

- Digital twins RT adaptability to deal with at run-time.
- Difficult interactions between the DT and the operators to make the right choice due diversity and big numbers of possible scenarios (e.g., robotics in dismantling nuclear plants).
- To create a high-performance and efficient digital twin model of a flexible/soft physical object such as tissues or cables.
- Modeling flexibility, friction, clearances, and the mechanical behavior of robots.

Physical Twin



Digital Twin



Technical stack

SysON ROS 2™ Rust